Folk dance as an ancient semiotic system and cultural replicator

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Research on the cultural evolution of human symbolic systems has focused mainly on language and music, while an equally ancient, complex, and globally distributed practice—folk dance—remains largely overlooked (Donald, 1993; Brown, 2000; Fitch, 2017). Far from mere entertainment, folk dance is a deeply embedded, culturally transmitted semiotic practice that encodes and sustains collective knowledge through structured, embodied form. Long before the advent of literacy, archaeological and ethnographic records—from 10-kya rock paintings (Dubey-Pathak, 2014) to ethnographic analyses of ritual movement (Hanna, 1987; Garfinkel, 1998)—attest that dance was among the earliest human symbolic activities, integral to cosmology, ritual, and social cohesion. As experimental studies of music (Verhoef & Ravignani, 2021; Anglada-Tort et al., 2023) and gesture (Motamedi et al., 2019; Fay et al., 2022) have shed light on symbolic evolution, incorporating dance can further illuminate how multimodal semiotic communication systems emerge, stabilize, and diversify across cultures.

We define folk dance as a symbolic system when it meets three criteria: (i) conventionalized, community-shared movement forms; (ii) public meanings anchored in cosmology or social role; and (iii) pragmatic force regulating ritual or emotional participation. These properties imply that dance, like language, is a learned, rule-governed, and meaning-bearing system shaped by transmission and selection. In cultural-evolutionary terms, folk dances function as *cultural replicators* (Pagel, 2009): patterns of embodied behaviour that preserve and transform shared knowledge through imitation and convention. They embody replicators as *motifs* (discrete kinematic units) and *combinators* (rules for sequencing, layering, and synchronizing movement). Maintained through pedagogy, ritual obligation,

and communal evaluation, movement forms achieve high fidelity while allowing variation and cultural adaptation. Their rhythmic synchrony, shared with vocal-learning species but not other primates (Coye et al., 2025) underscores their evolutionary depth as a means of social coordination and cohesion. Ethnographic examples from East-Asian traditions (Fig. 1) show how also socio-ecological factors shape these replicators by linking specific motifs, group formations and rhythmic patterns to myths or ecological knowledge (Hanna, 1987; Fink et al., 2021).

Figure 1. A: Han dances such as *Anhui Huagudeng* (安徽花鼓灯) retain the forward-leaning posture derived from the historical foot-binding custom **(B)**, which restricted gait for delicate, gliding motion. The small-step "lotus gait" encodes ideals of grace and restraint while revealing gendered socio-cultural values via bodily form. **C:** Mongolian dance styles embody the nomadic ecology: compact steps and vigorous shoulder vibrations reproduce the rhythm of horseback riding **(D)**. **E:** Tibetan dance, anchors meaning in cosmology. The lowered centre of gravity and clockwise circulation reflect adaptation to the high-plateau environment **(F)** and the Buddhist principle of *g.yas 'khor* (གཡས་སྐོར་)—movement aligned with the sun's course as an enactment of cosmic harmony **(G)**.

Dance movement systems can be analysed in ('super')linguistic terms (Patel-Grosz et al., 2023): A "kineme" inventory corresponds to features from Laban Movement Analysis (BODY, EFFORT, SPACE, SHAPE); combinators include sequence and parallelism; and style-specific constraints regulate allowable transitions and effort profiles. Many design features of language (Hockett, 1960) appear also in dance: *semanticity*, *arbitrariness*, *cultural transmission*, *compositionality*, and even *duality of patterning* (finite bodily configurations combine into rich symbolic repertoires). *Displacement* is manifested through ritual reenactment. Like grammar, these structures are compositional, e.g., Tibetan clockwise rotation indexes sacred alignment, while counter-rotation inverts that pragmatic frame.

We outline a future empirical research programme to investigate folk dance through an approach bridging ethnography, lab experiments, and computational modelling. Paradigms from iterated learning and experimental semiotics (Kirby et al., 2014; Tamariz, 2017) can be adapted to embodied signals, while motion-capture and pose-estimation enable quantitative study of movement structure, variation, and transmission fidelity. Modelling based on dynamical systems theory and dynamic semiotics can capture how local interactions among dancers generate stable, culturally specific attractors over time. Together, these approaches position folk dance as a co-evolving symbolic system—an embodied

archive of collective knowledge and a neglected window onto ancient communication systems that may have coincided with, or even preceded, the rise of symbolic vocal language.

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